



ISE 430/530: Manufacturing Systems and Facilities Design

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Facilities Planning Process

- Define products to be manufactured and/or assembled
- Specify required manufacturing and/or assembly processes and related activities
- Determine interrelationships among all activities
- Generate alternative facilities plans
- Evaluate the alternative facilities plans
- Select the preferred facilities plan
- Implement the facilities plan
- Maintain and adapt the facilities plan
- Update products to be manufactured and/or assembled and redefine the objective of the facility

Overview

Route sheet, assembly chart, operations process chart, precedence diagram

Determining the Production Requirements

- How many pieces are needed as inputs and outputs to each process, consider rework (if appropriate)
- Reject allowance (small volume)
- Number of required machines
- Machine assignment

Reject Allowance Problem

1. When producing small batches, the use of average values is less appropriate
2. Concerns
 - Cost, revenues, and probabilities associated with making good and bad products
3. Objective
 - Determine the number of additional units to allow when scheduling low-volume production where rejects occur randomly
4. For instance, consider a foundry that produces small numbers of custom-designed castings.
5. If the foundry has only one chance to produce the number of castings required, then the probability of a casting being good should be considered when determining the batch size to be produced.

Reject Allowance Problem

Determining the number of additional units to allow when scheduling low-volume production where rejects randomly occur is called the reject allowance problem.

x = the number of good units produced

$p(x)$ = probability of producing x good units

Q = quantity of units to produce

$C(Q, x)$ = cost of producing Q units, of which x are good units

$R(Q, x)$ = revenue from producing Q units, of which x are good units

$P(Q, x)$ = profit from producing Q units, of which x are good units

$$= R(Q, x) - C(Q, x)$$

$E[P(Q)]$ = expected profit from producing Q units

$$= \sum_{x=0}^Q P(Q, x) p(x)$$

The expected profit from producing Q units can be determined as follows:

$$E[P(Q)] = \sum_{x=0}^Q \{R(Q, x) - C(Q, x)\} p(x) \quad (2.4)$$

Reject Allowance Problem – Example:

An order for 20 custom-designed castings has been received. Each casting costs \$1100 to produce and has a recycle value of \$200 if unsold. The customer will pay \$2500 per casting for exactly 20 castings —no more, no less!. Based on historical data, Table 2.6 provides the probability distributions.

1. How many castings should be produced to maximize expected profit?

2. What is the probability of losing money at this production level?

Table 2.6 *Probability Distributions for the Number of Good Castings (x) out of Q*

# Good Castings (x)	Number of Castings Produced (Q)										
	20	21	22	23	24	25	26	27	28	29	30
12	0.05	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
13	0.05	0.05	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
14	0.05	0.05	0.05	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
15	0.05	0.05	0.05	0.05	0.00	0.00	0.00	0.00	0.00	0.00	0.00
16	0.10	0.05	0.05	0.05	0.05	0.00	0.00	0.00	0.00	0.00	0.00
17	0.10	0.10	0.05	0.05	0.05	0.05	0.00	0.00	0.00	0.00	0.00
18	0.15	0.10	0.10	0.05	0.05	0.05	0.05	0.00	0.00	0.00	0.00
19	0.20	0.15	0.10	0.10	0.05	0.05	0.05	0.05	0.00	0.00	0.00
20	0.25	0.20	0.15	0.10	0.10	0.05	0.05	0.05	0.05	0.00	0.00
21	0.00	0.25	0.20	0.15	0.10	0.10	0.05	0.05	0.05	0.05	0.00
22	0.00	0.00	0.25	0.20	0.15	0.10	0.10	0.05	0.05	0.05	0.05
23	0.00	0.00	0.00	0.25	0.20	0.15	0.10	0.10	0.05	0.05	0.05
24	0.00	0.00	0.00	0.00	0.25	0.20	0.15	0.10	0.10	0.05	0.05
25	0.00	0.00	0.00	0.00	0.00	0.25	0.20	0.15	0.10	0.10	0.05
26	0.00	0.00	0.00	0.00	0.00	0.00	0.25	0.20	0.15	0.10	0.10
27	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.25	0.20	0.15	0.10
28	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.25	0.20	0.15
29	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.25	0.20
30	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.25

Reject Allowance Problem – Example:

The revenue and cost functions can be given as follows:

$$R(Q, X) = \begin{cases} 200Q & x < 20 \\ 2500(20) + 200(Q - 20) & 20 \leq x \leq Q \end{cases}$$

$$C(Q, x) = 1,100Q \quad 0 \leq x \leq Q$$

$$P(Q, x) = \begin{cases} -900Q & x < 20 \\ 46,000 - 900Q & 20 \leq x \leq Q \end{cases}$$

Therefore, the expected profit can be given as follows:

$$E[P(Q)] = -\sum_{x=0}^{19} 900Qp(x) + \sum_{x=20}^Q (46,000 - 900Q)p(x)$$

The expected profit expression can be shown to reduce to

$$E[P(Q)] = -900Q + 46,000 \sum_{x=20}^Q p(x)$$

Plug in the numbers in Table 2.6

Table 2.7 Profit from Producing Q Castings, with Exactly x Being Good

# of Good Castings	Number of Castings Produced (Q)										
	20	21	22	23	24	25	26	27	28	29	30
12	−18,000	−18,900	−19,800	−20,700	−21,600	−22,500	−23,400	−24,300	−25,200	−26,100	−27,000
13	−18,000	−18,900	−19,800	−20,700	−21,600	−22,500	−23,400	−24,300	−25,200	−26,100	−27,000
14	−18,000	−18,900	−19,800	−20,700	−21,600	−22,500	−23,400	−24,300	−25,200	−26,100	−27,000
15	−18,000	−18,900	−19,800	−20,700	−21,600	−22,500	−23,400	−24,300	−25,200	−26,100	−27,000
16	−18,000	−18,900	−19,800	−20,700	−21,600	−22,500	−23,400	−24,300	−25,200	−26,100	−27,000
17	−18,000	−18,900	−19,800	−20,700	−21,600	−22,500	−23,400	−24,300	−25,200	−26,100	−27,000
18	−18,000	−18,900	−19,800	−20,700	−21,600	−22,500	−23,400	−24,300	−25,200	−26,100	−27,000
19	−18,000	−18,900	−19,800	−20,700	−21,600	−22,500	−23,400	−24,300	−25,200	−26,100	−27,000
20	28,000	27,100	26,200	25,300	24,400	23,500	22,600	21,700	20,800	19,900	19,000
21		27,100	26,200	25,300	24,400	23,500	22,600	21,700	20,800	19,900	19,000
22			26,200	25,300	24,400	23,500	22,600	21,700	20,800	19,900	19,000
23				25,300	24,400	23,500	22,600	21,700	20,800	19,900	19,000
24					24,400	23,500	22,600	21,700	20,800	19,900	19,000
25						23,500	22,600	21,700	20,800	19,900	19,000
26							22,600	21,700	20,800	19,900	19,000
27								21,700	20,800	19,900	19,000
28									20,800	19,900	19,000
29										19,900	19,000
30											19,000

Table 2.8 Expected Profit from Producing Q Castings

Number of Castings Produced (Q)										
20	21	22	23	24	25	26	27	28	29	30
−6,500	1,800	7,800	11,500	15,200	16,600	18,000	19,400	20,800	19,900	19,000

MAXIMUM



Number of Required Machines

Man & machine balancing

Machine (equipment) fraction

- Number of machines of one type required to produce the required volume of products
 - Some machines used for multiple operations
 - Some machines may not be able to perform the required operations fast enough
- Calculation based upon ALL operations the machine will perform



Number of Required Machines

The machine fraction for an operation is determined by dividing the total time required to perform the operation by the time available to complete the operation. The total time required to perform an operation is the product of the standard time for the operation and the number of times the operation is to be performed.

1. Are the parts actually being made to the 0.5 hour per part standard time?
2. Is the machine available when needed during the two-hour period?
3. Are the standard time, the number of parts, and the time the machine takes known with certainty and fixed over time?

$$F = \frac{SQ}{EHR}$$

Standard time per unit

Number of units

Machine Fraction

Performance (% time)

Reliability of machine, expressed as percent uptime

Time available per machine

Number of Required Machines - Example

See Eq. 2.5 and Example 2.5

- Standard machining time of 2.8 minutes per part, 200 units needed in 8-hour shift, milling machine will be operating 80% of time. During operational time, parts are produced at 95% rate

$$F = \frac{SQ}{EHR} = \frac{2.8\text{min} * 200}{480\text{min} * 0.8 * 0.95} = 1.54 \text{ machines needed}$$

Number of Required Machines

Combine requires for identical machine types

Table 2.9 *Total Equipment Requirement Specification Example*

Operation Number	Equipment Fraction	Next Highest Whole Number
109	1.1	2
206	2.3	3
274	0.6	1

It may be seen that a minimum of four and a maximum of six machines are required.

With no further information, a specific recommendation on the number of required machines cannot be made.

How many machines are needed?

Unclear, need more information

Job Shop/Production Line

Cost, Set-ups, Inventory, etc.

Future growth

Overview

Determining the Production Requirements

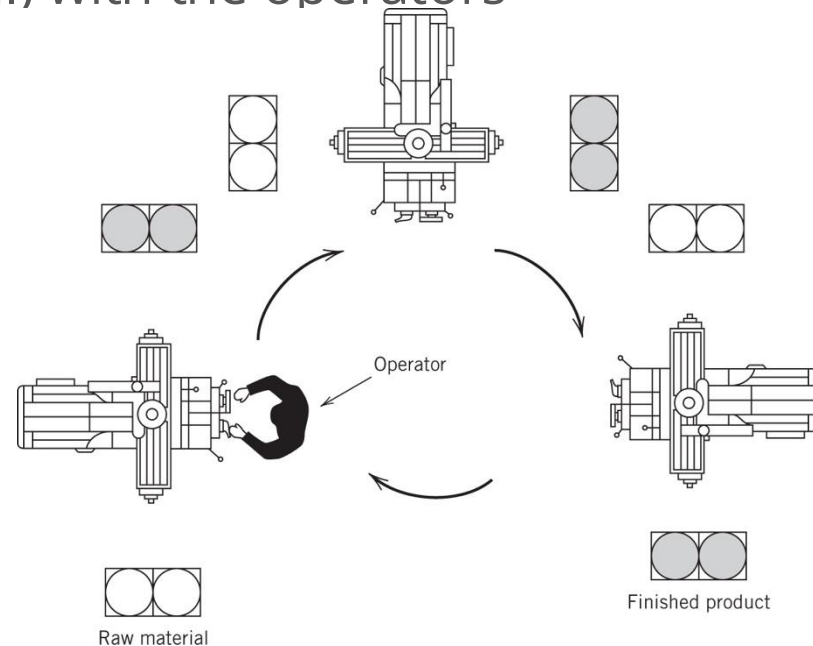
- How many pieces are needed as inputs and outputs to each process, consider rework (if appropriate)
- Reject allowance (small volume)
- Number of required machines
- Machine assignment

Machine Assignment Problems

Considers the assignment of machines (semi-automated) to operators

Human-machine chart (aka. multiple activity chart)

- Relationship of multiple activities and machines (non-identical) with the operators



Time	Operator	Machine 1	Machine 2	Machine 3			
0	L-1	Loaded	Idle	Idle	Transient State (12 min)		
1	T-2	Machining	Idle				
2	L-2		Loaded				
3	T-3		Machining			Loaded	
4	L-3						
5	T-1						
6	Idle		Machining	Machining			
7	UL-1	Unloaded					
8	L-1	Loaded					
9	I&P-1	Machining	Idle	Idle	Steady-State Repeating Cycle (9 min)		
10	T-2		Unloaded				
11	UL-2		Loaded				
12	L-2		Machining			Machining	
13	I&P-2						
14	T-3						
15	UL-3	Unloaded					
16	L-3	Loaded					
17	I&P-3	Idle	Machining	Machining			
18	T-1	Machining					
19	UL-1					Unloaded	
20	L-1					Loaded	
21	I&P-1					Machining	Machining
22	T-2						
23	UL-2						
24	L-2						

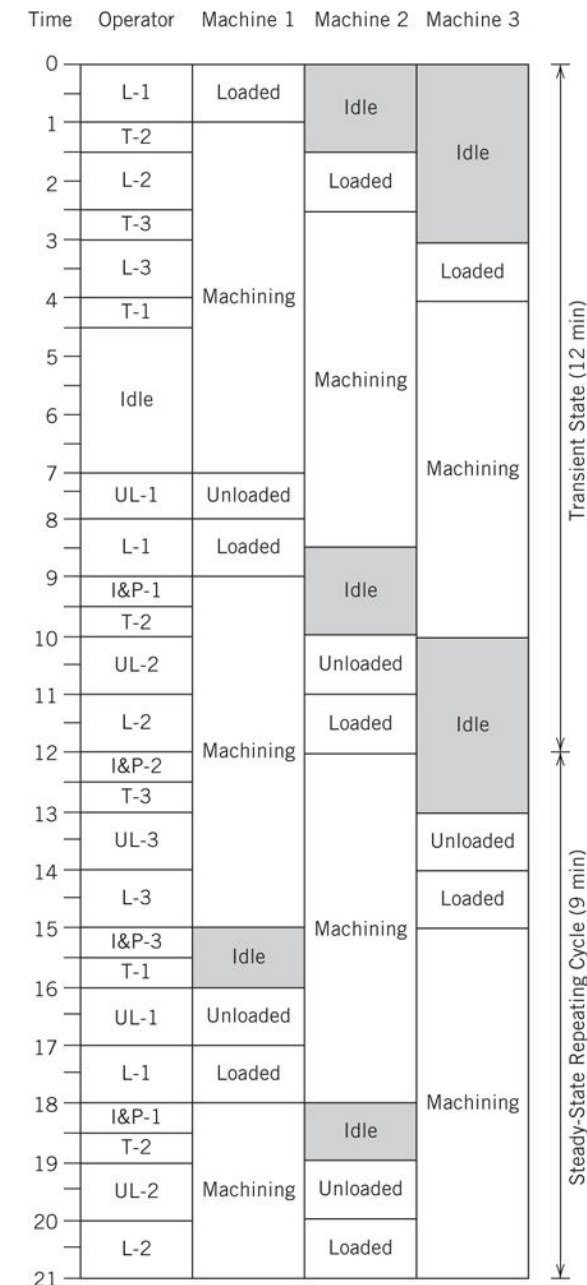
Machine Assignment Problems

Deterministic model (Example 2.7)

For the situation illustrated in Figure 2.18, it takes 0.5 minute to travel between machines, 1 minute to load a machine, 1 minute to unload a machine, 6 minutes of automatic machine time, and 0.5 minute to inspect and pack a finished part. As shown, the analysis begins with each machine empty and the operator standing in front of Machine 1 (M-1).

The operator loads M-1, walks to M-2, loads M-2, walks to M-3, loads M-3, walks to M-1, unloads M-1, loads M-1, inspects and packs the part removed from M-1, travels to M-2, and so forth.

As shown in Figure 2.18, it takes 12 minutes for the operator and three machines to achieve a steady-state condition; thereafter, a repeating cycle of 9 minutes in duration occurs. (In other words, if nothing interrupts the activities of the operator and the three machines, the 9-minute cycle will repeat indefinitely.)



Machine Assignment Problems

a = concurrent activity time (man and machine)

b = independent operator time

t = independent machine time

n' = ideal number of identical machines to assign an operator

m = number of identical machines assigned

T_c = repeating cycle time

I_o = idle operator time during a cycle

I_m = idle time for each machine during a cycle

Excluding idle time, each machine cycle requires $a + t$ minutes to complete a cycle. Likewise, the operator devotes $a + b$ minutes to each machine during a cycle. Hence, an ideal assignment is

$$n' = (a + t)/(a + b) \quad (2.6)$$

For the example, a equals two minutes, b equals one minute, and t equals six minutes. Therefore, n' equals 2.67 machines.

2 would cause the operator idle time

3 would cause the machine to have idle time

Machine Assignment Problems

Since a fractional number of machines cannot be assigned to an operator, consider what will happen if some integer number of machines, m , is assigned. The work content for the operator will total $m(a + b)$, while a machine cycle will be $(a + t)$ in duration. The repeating cycle will be the larger of the two, and the difference in the two will be idle time. If $m > n'$, then the repeating cycle will be $m(a + b)$ in duration; if $m < n'$, then the repeating cycle will be $(a + t)$ in duration. Therefore,

$$T_c = \begin{cases} (a + b) & m \leq n' \\ m(a + b) & m > n' \end{cases} \quad (2.7)$$

$$I_m = \begin{cases} 0 & m \leq n' \\ T_c - (a + t) & m > n' \end{cases} \quad (2.8)$$

$$I_o = \begin{cases} T_c - m(a + b) & m \leq n' \\ 0 & m > n' \end{cases} \quad (2.9)$$

For the example, with $m = 3$ and $n' = 2.67$, the repeating cycle is $3(2+1)$, or nine minutes as observed in Figure 2.18. The idle time for each machine during a repeating cycle equals $9 - (2+6)$, or one minute, again as observed in Figure 2.18. As shown in Figure 2.19, the work cell must be designed carefully to provide adequate access for incoming and outgoing material, machine maintenance, and operator movement.

You want to find out how many CNC Machines to assign to each operator. Time to unload one part and load the next part is 3 minutes. During this time the machine is stopped. It takes 20 minutes to run a part. The operator must inspect each part which takes 5 minutes. The inspection is done on the work bench. It takes 0.5 minutes to travel between machines.

Calculate the total cost for 2 and for 3 machines find the lower cost (min Tc):

For $m < n'$: $T_c(m) = (C_o + mC_m)(a+t)/m$

For $m > n'$: $T_c(m) = (C_o + mC_m)(a+b)$

The cost to run the machine is \$55/hr

The cost of an employee is \$40/hr

2 machines per employee:

$m=2$

$TC = (C_o + mC_m)(a+t)/m = (40+255)*(3+20) / 2. = 1725$ • min at $m=2$

3 machines per employee

$m = 3$

$TC = (C_o + mC_m)(a+b) = (40+355)(3+5.5) = 1745.5$